Bayesian Statistics II — Hierarchical Models

ECON 8310: Business Forecasting — Lecture 11, Part 2

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Lecture Outline

1 Which Series May Be Pooled

2 Partial Pooling

3 Fitting It, and the Trap

4 Reading the Result

5 Key Takeaways

Which Series May Be Pooled

Before you pool anything, ask what the groups have in common.

The Question

Does demand rise in weeks when SNAP benefits are disbursed, and by how much — store by store? Homework 3's memo stocked up store by store, assuming those differences are real. Today we check.

One question comes before the model. SNAP is a *food* assistance program. The raw SNAP-week effect on log units, all thirty series by category:

Category	Mean effect	Spread across stores
FOODS (10 series)	$+0.089$	0.058
HOBBIES (10 series)	$+0.006$	0.010
HOUSEHOLD (10 series)	$+0.011$	0.014

These are not thirty versions of the same thing. FOODS responds; the other two barely move. Pooling all thirty would tell the model they are interchangeable.

Exchangeability: the Assumption Under Every Hierarchy

Exchangeable

Groups are **exchangeable** when, before seeing the data, you have no reason to expect one to differ from another in a particular direction. Swapping their labels would not change what you believe.

The ten FOODS stores qualify: nobody could say in advance which store's SNAP effect is largest. FOODS against HOBBIES does **not**: a food program moves food.

Today: the **ten FOODS series**, five thinned to **8 weeks**. That imbalance is the point.

The lab pools all thirty anyway, and measures what it costs.

Complete Pooling and No Pooling

Within that exchangeable set there are two obvious approaches, and each fails in a different direction.

	What it does	How it fails
Complete pooling	One SNAP effect for all ten series. `b = pm.Normal("b", 0, 1)`	Denies that stores differ at all. A genuinely different store is averaged away.
No pooling	A separate effect per series, estimated independently. `shape=10`	A series with 8 weeks gets an estimate built from 8 weeks. Mostly noise, reported as a finding.

You have used both already without naming them. Homework 3's random forest pooled all thirty series into one model with store dummies — close to complete pooling. Homework 1 fitted ARIMA to one series at a time — no pooling. The choice has always been made by convenience; partial pooling makes it a modeling decision and lets the data settle it.

Partial Pooling

Let each series have its own effect — drawn from a shared distribution the data also estimates.

The Hierarchy Is One Extra Line

Give each series its own effect b_j, but say where those effects come from:

$$b_j \sim \mathcal{N}(\mu_b,\ \sigma_b), \qquad j = 1, \ldots, 10$$

μ_b is the typical SNAP effect across FOODS, and σ_b is **how much stores genuinely differ**. Crucially, both are estimated from the data rather than assumed.

$\sigma_b \to 0$	The model concludes stores do not differ, and reproduces complete pooling.
σ_b large	The model concludes they differ a lot, and approaches no pooling.
In between	Each b_j is pulled toward μ_b — *how far* depending on how much data that series has.

That last row is the whole idea. A series with 277 weeks barely moves; one with 8 weeks moves a long way. Nobody chose those weights — they fall out of the model.

Why This Is the Right Answer, Not a Compromise

Partial pooling sounds like splitting the difference. It is better than that.

With 8 observations, a series' own estimate is mostly noise. Pulling it toward μ_b is the right response, not a fudge: you would not call a coin biased after three flips.

Regularization, for the sixth time. The prior shrinks each group toward μ_b as Ridge shrinks coefficients toward zero — but the strength, σ_b, is *estimated*, not tuned by cross-validation.

Smoothing penalty (L3) → pruning (L4) → `reg_lambda` (L5) → Ridge and LASSO (L6) → weight decay (L7) → hierarchical priors.

Fitting It, and the Trap

The obvious code does not work, and the reason is geometry rather than statistics.

The Model in PyMC

Non-centered hierarchical model

```
with pm.Model() as model:

mu_b = pm.Normal("mu_b", 0, 0.5)       # typical SNAP effect

sd_b = pm.HalfNormal("sd_b", 0.25)     # how much stores differ

z    = pm.Normal("z", 0, 1, shape=J)    # non-centered

b    = pm.Deterministic("b", mu_b + z * sd_b)

a = pm.Normal("a", y.mean(), 1, shape=J)  # per-series level

s = pm.Exponential("s", 1.0)

pm.Normal("obs", mu=a[idx] + b[idx]*snap, sigma=s, observed=y)
```

`idx` maps each row to its series, so `b[idx]` picks out the right effect for every observation. That indexing pattern is how every hierarchical model in PyMC is written.

Note what is *not* here. No decision about how much to pool, no weighting scheme, no threshold for "enough data." Those all come out of estimating σ_b.

The Trap: Write It the Obvious Way and It Breaks

The natural way to write the hierarchy is `b = pm.Normal("b", mu_b, sd_b, shape=J)` — the *centered* form, which reads exactly like the equation. On our data it produces:

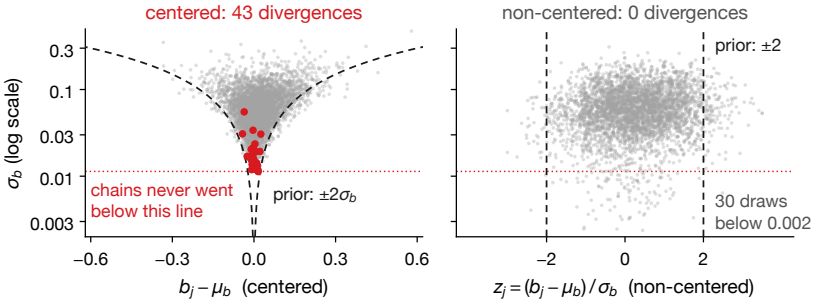
	Centered	**Non-centered**	**Want**
Divergences	**43**	**0**	0
$\hat{R}$	1.003	1.003	< 1.01
ESS (bulk)	966	1,719	> 400

Only one alarm fires. $\hat{R}$ and ESS pass; on those alone you would ship it. **43 divergences against a threshold of zero** is the objection to trust.

Counts move with the seed, even the machine. Ten more seeds: 15–285 divergences centered, 0–11 non-centered; $\hat{R}$ or ESS objected in 2 of the 11 runs, divergences in all 11.

Why It Breaks: a Funnel

Every posterior draw for one thin series (CA_1), against σ_b on a log axis:



Given σ_b, the prior spreads $b_j - \mu_b$ by σ_b: a funnel, pinched where σ_b is small. The divergences sit in the neck (39 of 43 in the bottom tenth of σ_b), and the chains go no lower. $b_j = \mu_b + \sigma_b z_j$ is a change of coordinates: z_j spreads by 1 at every height, a column the sampler can walk. Exact for the prior; the posterior follows closely.

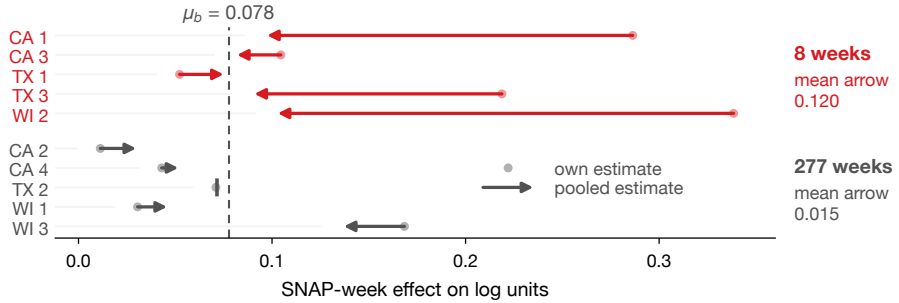
BMCP §4.6.1, Posterior Geometry Matters.

Reading the Result

Shrinkage, measured — and checked against data the model never saw.

Shrinkage, Measured

Each series' own (unpooled) estimate, and where the hierarchy moved it:

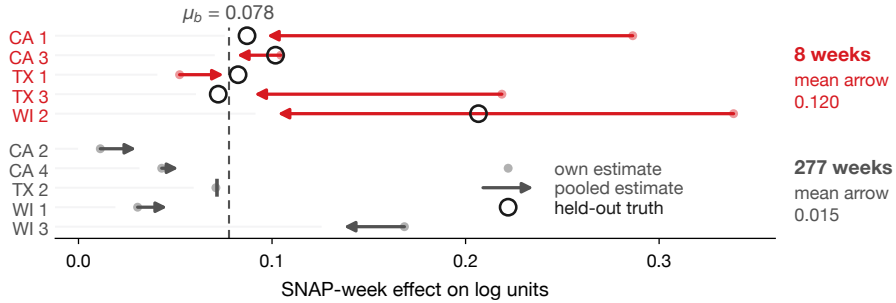


Thin series moved 8.1 times further toward μ_b, though nobody said which were thin. Full series keep most of their spread ($0.055 \to 0.037$); thin ones lose it ($0.108 \to 0.010$).

Each pooled estimate sits approximately at a precision-weighted point between μ_b and its own: thin series keep about a tenth of theirs, full series about three-quarters. Only approximately, because μ_b and σ_b are integrated over.

Checked Against Data the Model Never Saw

The weeks we thinned away still exist. Each ring is the SNAP effect observed in them:



RMSE against the truth: 0.126 unpooled, 0.048 pooled — 2.6 times smaller. Pooling won on four series of five; WI_2 shows the cost, pulled past its truth.

Borrowing strength is not a figure of speech: the thin series were estimated better because the other five existed.

What the Model Actually Concluded

The estimated hierarchy: $\mu_b = 0.078$ (94% HDI about $[0.02,\ 0.15]$) and $\sigma_b = 0.065$.

So a SNAP week lifts FOODS units by about **8%**, and stores genuinely differ around that by about **7 percentage points**.

> **Not a license for store-by-store stocking.** The thin stores' own estimates ran from 5% to 40%. The held-out weeks say that was mostly noise; $\sigma_b = 0.065$ says not *all* of it.

The defensible plan: 8% everywhere, modest store variation around it, and no store's own estimate until it has the weeks to support one.

Key Takeaways

What hierarchy buys, and what it costs.

Lecture 11 Part 2: Key Takeaways

1. **Ask what may be pooled first:** groups must be **exchangeable**. FOODS and HOBBIES are not.
2. **Complete pooling** denies groups differ; **no pooling** reports noise as a finding.
3. **Partial pooling** draws each group's parameter from a shared distribution whose spread is estimated.
4. **Shrinkage is proportionate:** thin series moved **8.1**$\times$ further; held-out error was **2.6**$\times$ smaller.
5. **Regularization for the sixth time**, with the strength estimated, not tuned.
6. **Write it non-centered.** 43 divergences became 0: same model, a change of coordinates.

Next: Bayesian linear regression — coefficients as distributions.

References I